

Li Lab Guidebook

University of Pennsylvania

RNA, Genetics, Immune

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Welcome

Welcome to the Li lab at the University of Pennsylvania – we are excited to have you!

We are a group interested in the RNA-mediated immune regulation and study the relevant processes and diseases using methods from the fields of genetics/genomics, molecular and cellular biology, and immunology.

During your time in the lab, we hope that you do at least four things:

- 1) Have fun and make new friends
- 2) Develop new and useful skills
- 3) Collect and explore interesting and exciting data
- 4) Learn something about RNA/genetics/immunology

This lab manual was inspired by several other lab manuals and builds upon some of the content from them (e.g. the [Aly Lab manual](#) and the [MemoLab manual](#)).

Upon joining the lab you are expected to read the sections of manual relevant to you.

Expectations and Responsibilities

Everyone

In order to maintain a work environment that is inclusive, engaging, safe, welcoming and rewarding there are several things we all must do:

The 'big' things:

- Support your lab mates. Help them out (even if you are not 'on the project') and be a friendly citizen when they need it. Science should be collaborative, not competitive. If you help others, you can expect others to help you when you need it.
- Respect your lab mates. Respect their personal space and preferences. Respect their culture, religion, beliefs, and sexual orientation. Everyone should feel welcome in the lab and discrimination or harassment of **any kind** will not be tolerated.
- If you are struggling, please tell someone. Science can be hard, and we are here to help. Please don't suffer alone.
- Work carefully. Rushing an experiment/analysis often leads to the wasting of resources and/or making mistakes. Mistakes happen – but they should not be because of carelessness.
- If a mistake is made, admit it, correct it, and move on. This includes telling collaborators after you have already shown results to them or they are already working based on the data from where an error has been discovered.
- Honesty is the best policy, especially in science. It is never ok to plagiarize, tamper with data, fabricate data, or fudge results. Null or unexpected results are normal and still important. We can do important, cool and high impact science without resorting to these practices. This cannot be emphasized enough; **academic misconduct will not be tolerated**.

- If there is tension, conflict or hostility in or outside of the lab, it needs to be resolved as soon as possible. If you don't feel comfortable confronting the person in question or if the situation is escalating, tell Qin.
- If you have a problem with Qin and are comfortable telling him about it, please do so. If you aren't comfortable, then tell another lab member, or another member of the department, or another senior faculty member/mentor (for more serious issues).

The 'little' things that keep things running smoothly:

- Keep the lab, and common areas uncluttered.
- Contribute in lab meeting and come prepared – especially if you are the primary presenter.
- Show up to scheduled meetings and be on time.
- Respect people's time – attempt to answer/solve your own questions or problems before asking someone else.
- If you see something is not working, running low or empty either do something about it yourself or find someone else to take care of it – don't just 'pass the buck'.

Principal Investigator

All of the above, and I will commit to:

- Supporting you scientifically, emotionally, & financially.
- Provide feedback on your work (ideas, posters, talks, manuscripts etc) in a timely manner.
- Be consistently available to you for in person meetings and via e-mail/slack.
- Support your career development by introducing you to other researchers in the field, promoting (and crediting) your work in talks, writing recommendation letters, and letting you attend conferences as finances permit.
- Help you to prepare for the next step of your career (inside or outside of academia).
- Maintain an environment that facilitates physical and emotional well-being.

Postdoctoral Fellows

In addition to the above for everyone, you are also expected to:

- Develop and make progress in your own independent line of research.
- Help to train and mentor students in the lab (undergraduate and graduate) when needed.
- Present your work at events within and outside of the university (e.g. national and international conferences, seminar series, departmental events).
- Apply for funding. This is part of your training for moving to the next phase of your career. It is also in your best interest to gain experience writing grants and can help to extend your contract and/or free up lab funds for other uses.
- Challenge Qin when he is wrong or when your opinion is different.
- Attend relevant departmental seminars whenever possible.

- Initiate and build collaborations, either within or outside of the university.

Graduate Students

In addition to the above for everyone, you are also expected to:

- Develop your dissertation research projects – for PhD students this should be at least 2 substantial projects that address a clearly defined question.
- Present your work at events within and outside of the university (e.g. national and international conferences, seminar series, departmental events).
- Apply for internal and external funding to support your research. It is useful experience applying for small grants and awards, looks good on the CV, and can give you more project freedom.
- Help to mentor undergraduate students in the lab as needed.
- Prioritize your time towards research. Coursework is important, but research is the core contribution towards your PhD.
- Ensure that you are aware of all departmental deadlines and degree requirements.
- Attend relevant departmental seminars whenever possible.

Lab Manager / Research Technicians

In addition to the above for everyone, you are also expected to:

- Work on any of your own research projects (developed with Qin's help).
- Maintain the lab facility.
- Help new lab members settle into the lab by answering whatever questions they have that you can answer or directing them to Qin as appropriate.
- Oversee hiring, scheduling and training of undergraduate research assistants.
- Maintain the lab website, update the lab manual, add events to the lab calendar.
- Assist lab members with data collection as necessary.

Undergraduate Students

In addition to the above for everyone, you are also expected to:

- Develop a weekly schedule to spend time in the lab by talking to your mentor. You should be coming in every week and scheduling enough time to get your work done.
- Assist other lab members with data collection and analysis (unless you are working on your own independent project under the mentorship of another lab member, in which case you should work on that).

Code of Conduct

The lab, and the university, is an environment that must be free of harassment and discrimination. All members are expected to abide by the University of Pennsylvania policies on discrimination and harassment, which you can (and should) read: the [Code of Student Conduct](#) and the [Sexual Violence and Sexual Harassment Policy](#).

Guidelines, General Policies and Lab Events

Authorship

There are two key requirements for being an author on a manuscript:

- 1) Contribute to the intellectual content of the manuscript in a meaningful way.
 - 2) Contribute to the writing and/or editing of the manuscript in a meaningful way.
- Note that “collect data”, “analyze data”, or “fund the study” are *not* on the list. Those are key components of a paper, but do not (on their own) necessarily warrant authorship.
 - Most often one person will take on the main responsibility for writing the paper, and this person will be given first authorship.
 - It is worth noting that there are countless opinions regarding authorship, and there are always borderline cases. When it comes to borderline cases, I generally suggest erring on the side of generosity. When collaborating with other people, I tend to defer to their lab group’s criteria. However, it’s important that within our group, we are clear on the expectations for authorship and transparent from the beginning about authorship discussions and decisions. If you ever have any questions or issues, please come speak to me.

Celebrations

- **Birthdays** will be celebrated with baked goods (or alternative treats if so desired) – so don’t forget to put that all important date in the lab calendar!
- **Publications.** Publications are something to be celebrated! If you publish a paper, it is the lab’s tradition that you provide a beverage of your choice (alcoholic or not) to share with your lab mates.
- **Major life events** (e.g., engagements, weddings, new babies, graduations, or landing a new job!) Your life outside the lab is just as important as your work inside it. For these significant milestones, the lab will typically organize a group card, a small gift, or a lab lunch/happy hour to celebrate together. If you have a milestone coming up that you’d like to share, please let Qin know so we can plan accordingly! (And if you prefer not to make a fuss, just let us know that, too).

Deadlines and Feedback

- If something is important and has a firm deadline, it is crucial to tell your collaborators and people whose help you need. You should tell the relevant people about the deadline as soon as you know when it is. Please also make sure to remind them as the date approaches. This also means you shouldn’t be afraid to ‘bug’ someone about it.
- Give Qin at least **one week’s** notice to do something with a hard deadline that doesn’t require a lot of time (e.g. reading a conference abstract, filling out paperwork).
- Give Qin at least **two weeks’** notice for something that will take a moderate amount of time (e.g. letter of recommendation, award nomination).
- If you want feedback on something that will require multiple back-and-forth interactions between you and Qin (research statement, teaching statement, grant application) it would be best to start this process at least **four weeks** before the submission deadline.

- For manuscript submission and revisions (*i.e.* when there is no hard deadline), send drafts to Qin as soon as you have them and bug him to give you feedback if two weeks have passed.

Hours

- There are no precise hours you are expected to keep. As long as you are getting your work done you can do it at whatever time of day you like. One of the benefits of academia is having flexibility in your work schedule.
- In this vein, you are welcome to work 'off campus' when desired (*i.e.* from home, a coffee shop *etc.*).
- You are however expected to maintain a regular and meaningful overlap on a daily basis with other members in the lab.

Lab Calendar

- The following should be added to the lab calendar:
 - Planned absences for: extended vacations, conference travel.
 - Your birthday!
 - Lab recreational events.

Lab Duties

- Keeping the lab running is a shared responsibility. In addition to the 'little things' listed above, every full-time lab member (grad students, postdocs, technicians) is assigned one or more communal lab duties. Undergraduates may be assigned lighter duties in consultation with their mentor.
- Duties are assigned at the beginning of each semester and rotate so that nobody is stuck with the same job indefinitely. The current assignments are posted in the lab and pinned in **#general** on Slack.
- Typical duties include:
 - Preparing and restocking common solutions and shared reagents (*e.g.* buffers, 70% ethanol, LB and agar plates, RNase-free water).
 - Autoclaving, glassware washing, and restocking tips, tubes and other consumables in the common areas.
 - HPC lab space monitoring: keep track of the lab space usage on PARCC and PMACS, remind other lab members to clean up their space and properly backup data, maintain the code repositories tidy.
 - Tissue culture room upkeep: media stocks, incubator water pans and CO₂ tanks, cleaning the biosafety cabinets, and biohazard waste.
 - Freezer and cryo-storage upkeep: checking –80 °C freezers and liquid nitrogen levels, defrosting the –20 °C freezers, and keeping the freezer inventory current.
 - Chemical and biological waste: labeling waste containers, filling out hazardous waste tags and scheduling EHRS pickups.

- Shared equipment care: cleaning centrifuges, water baths and gel boxes; checking pipette calibration schedules; keeping the equipment sign-up binders stocked.
- Ordering point of contact for a given category of supplies (see Ordering and Lab Stocks).
- Do your duty on schedule and without needing to be reminded. If you will be away (travel, vacation, illness), arrange for someone to cover and let the lab manager know.
- Duties should take no more than 1–2 hours a week. If a duty is regularly taking much longer than that, or if the workload is uneven, tell the lab manager or Qin so we can rebalance.
- Lab duties are not a substitute for everyday courtesies: if you empty something, notice something broken, or make a mess, fix it yourself regardless of whose duty it is.

Lab Equipment

- Treat equipment with respect.
- During busy times sign up to use ‘in demand’ items (qPCR machine *etc.*). This is done using the binders adjacent to each piece of equipment to avoid disappointment/double booking.
- If something isn’t working, let the lab manager know so we can get it fixed.

Lab Notebook

- Every lab member doing bench or computational work must keep a lab notebook. Your notebook is the permanent record of what you did, the basis for reproducing your results, and what we rely on when writing up methods, responding to reviewers, or investigating a discrepancy years later.
- You may use a bound paper notebook or an electronic lab notebook (ELN, such as Benchling). Pick one and use it consistently. Whichever you choose, it must be accessible to Qin and, for electronic notebooks, backed up in at least one other location (see Reproducibility and Data Management).
- For each experiment, record at minimum:
 - The date and a short title, plus a page/entry number that you can reference from your sample labels and data files.
 - The aim or hypothesis – one or two sentences on why you are doing the experiment.
 - The protocol you *actually* followed, including any deviations from the standard protocol, volumes, times, temperatures, and instrument settings.
 - Reagent details: catalog and lot numbers, antibody clones and dilutions, plasmid and primer names, cell line passage numbers, mouse IDs and genotypes.
 - Sample IDs and where the samples are stored (freezer, rack, box, position).
 - Raw data (or, for large files, the exact path to where they live on the server) and any figures, gel images or plots.
 - A brief interpretation and the next steps – including when the experiment failed. Negative and failed results are data and must be recorded too.

- Write entries the **same day** you do the work. Do not reconstruct experiments from memory days later, and never write up ‘expected’ results before the experiment is finished.
- If using a paper notebook: use pen, number the pages, never remove pages or use correction fluid, cross out mistakes with a single line (so they remain legible) and initial/date the change, tape or glue in printouts, and keep a running table of contents at the front.
- For computational work, your ‘notebook’ should capture what a paper notebook would: the question, the input data and its provenance, the code version (*e.g.* git commit), software and package versions, parameters used, and a summary of results. A dated markdown/README file alongside a version-controlled analysis directory is acceptable if it is cross-referenced from your main notebook.
- Write for someone else. The test is whether another lab member could pick up your notebook and repeat the experiment without asking you any questions.
- Qin may ask to look at your notebook during individual meetings, and will do so when preparing manuscripts. Notebooks (paper and electronic) belong to the lab and the University and stay in the lab when you leave; you are welcome to take a copy.

Meetings

Individual Meetings

- At the beginning of each semester, we will set a schedule for weekly meetings. Each full-time lab member (grad students, postdocs, technician) will have a regular 1-on-1 meeting with Qin. These meetings can be used however you find most valuable; this can include looking at data, discussing ideas, working through problems with an analysis, talking about future plans *etc.* If you feel there is nothing or little to discuss you are welcome to cancel the meeting or just drop in for a quick chat.
- Qin will meet with undergraduate students every other week (or according to need) – but post-docs and graduate students should meet with their undergraduate mentee on a more frequent basis.

Lab Meetings

- Two-hour long Lab Meetings are held once a week at TBD time in TBD location.
- Lab members will take turns leading. A schedule assigning weeks will be created at the beginning of each semester.
- When it is your week to present you typically have the options of:
 - Getting feedback on a manuscript you are working on.
 - Presenting an update on an in-progress or completed project.
 - Leading the discussion of a journal article or topic of your choice.
 - Giving a practice talk for an upcoming seminar, conference presentation or job talk.
- If leading the discussion of a journal article, or if it is your turn to lead an assigned journal club, the paper should be sent to the lab email list at least 7 days before lab meeting to give adequate time for reading.

Wet Lab Cleanliness and Safety

- Common use areas should be kept clean and clutter free.
- No food should be in the molecular lab.
- All bottles containing liquids should be properly labeled.

Ordering and Lab Stocks

- If you use the last of something or notice stocks are low for a commonly used item, order it and notify the lab manager so that it can be reordered.
- For non-routine restock orders or orders over \$500 USD, notify the lab manager or ask Qin for approval before placing the order.

PI Office Hours

- In addition to weekly meetings, you can most often find Qin in his office. Feel free to swing by for a chat, but please knock first.
- Recommendation Letters. If you have been in the lab for more than six months, you can count on Qin to write you a letter. Exceptions can be made if students or post-docs are applying for fellowships shortly after starting in the lab.
- If you need a letter, notify Qin of the deadline as soon as possible (see Deadlines and Feedback for more guidance). Also send Qin your CV, and any relevant instructions for the content of the letter (*e.g.* what exactly it is for and where to send it). If the letter is for a faculty position, also include your teaching and research statements.

Reproducibility and Data Management

- In order to have confidence in our work, it is important that others are able to reproduce our results. Reproducible research is an essential component of science and an expectation for all projects coming out of the lab.
- To ensure reproducibility, your analysis pipeline should be well organized and well documented. It is therefore important to take extensive notes for each step of your analysis pipeline – this includes indicating any pre-processing of the data. It is also important to keep detailed notes on your experimental design. Your code should be clearly commented and written in a way that would be interpretable to someone else.
- All data, code and samples must be archived. For now, you can use your preferred ways/methods for archive, but please make sure they are accessible to Qin.
- All raw sequence data should be stored in *at least* two places at any time.

Safety in the Lab

- [Safety Data Sheets](#)
- [Chemical Hygiene Plan](#)
- Laboratory Safety Manual – policies and procedures for a variety of things: [PennEHRS Laboratory Safety](#)

Sick Days

- If you are sick stay at home unless it is absolutely 100% necessary to come in to work (e.g. an entire experiment will be ruined because you can't find someone to cover for you). Nobody wants your germs. Take the time to rest or, if you are feeling well enough to work, find something you can do at home – writing, catching up on the literature; the list is usually endless...

Travel

- PhD students who are presenting at a conference are eligible to receive \$1000 of travel support per fiscal year (July 1 – June 30) from [BGS](#). I will provide at least an additional \$500 per year towards the costs of conference related travel (maximum is funding and situation dependent).
- Post-doctoral fellows will receive at least \$1500 per year for conference travel (maximum is funding and situation dependent).

Vacation

- Let Qin know if you are intending to be away on vacation. Vacations under 2 weeks don't need 'pre-approval' from Qin. Vacation durations > 2 consecutive weeks should be discussed – ideally before plans are made.

Lab Resources and Communication Channels

Slack

Slack will be used as the primary means of lab communication.

Notes on using the lab Slack:

When posting messages or looking for updates, check the appropriate channel:

- **#general** – lab announcements
- **#lab-meetings** – schedules, notes or communication related to lab meetings
- **#papers** – sharing links to relevant papers and discussing them
- **#seminars** – sharing seminar flyers and discussing talks
- **#random** – non-work-related chatting that is best kept out of the work-related channels

Full-time lab members should install Slack on their computers and/or phones. Part-time lab members should also check Slack regularly. You should of course feel free to ignore Slack on evenings and weekends.

Google Calendars

The Li Lab calendar: used to keep track of lab events, birthdays and travel. (Link TBD)

To be continued...